



Cardiac: Narrow Complex Tachycardia with Pulse

Assessment

Pediatric Pearls:

- Use pediatric dosing of medications or electrical therapy for a pediatric patient < 37 kg and as defined by HandTevy
- Focus on rapid and early BLS airway and ventilation tools. Intubation may not be the best option for these patients.
- Pediatric pads should be used in children < 10 Kg or HandTevy color purple.

Signs & Symptoms:

- QRS $\leq$ 0.12 sec
- Pale or Cyanosis
- Diaphoresis
- Tachypnea
- Vomiting
- Hypotension
- Altered Level of Consciousness
- Pulmonary Congestion
- Syncope

Differential:

- Heart disease (WPW, Valvular)
- Sick sinus syndrome
- Myocardial infarction
- Electrolyte imbalance
- Exertion, pain, emotional stress
- Fever
- Hypoxia or Anemia
- Hypovolemia
- Drug effect / Overdose
- Hyperthyroidism
- Pulmonary embolus
- Alcohol withdrawal

Clinical Management Options

P	P	P	P	P	P		• Oxygen PRN titrated to SpO₂ 92%-96% • Basic airway management
L	L	L	L	L	L		
1	2	3	4	5	6		• 4 lead and 12 lead ECG placement and acquisition • Apply EtCO₂
							• Vascular access • Isotonic Crystalloid PRN titrated to SBP $\geq$ 90 mmHg or MAP $\geq$ 65
							• Monitor and interpret of ECG • Valsalva Maneuver • Adenosine x2 doses as needed • Continuous 12-lead ECG during Adenosine administration, if possible • Diltiazem (Adults only) – DO NOT admin if Hx or 12 lead presentation of WPW ◦ If there is a clinical concern with Diltiazem, Amiodarone is the alternative. • Sedation: Midazolam or Ketamine as appropriate • Adult Synchronized Cardioversion at maximum joules • Pediatric Synchronized Cardioversion 0.5 to 1.0 j/kg, repeat as needed at 2 j/kg • 12 lead ECG post-conversion

Consult Online Medical Control As Needed

PEDIATRIC CARIOVERSION DOSE CHART

Preemie	Newborn	4MO	6MO	1YR	2YR	3YR	4YR	5YR	6YR	7YR	8YR	9YR	10YR	11YR	12YR	13YR
2KG	4KG	6KG	8KG	10KG	12KG	15KG	17KG	20KG	22KG	25KG	27KG	30KG	35KG	40KG	50KG	60KG
1j	2j	3j	4j	5j	6j	8j	9j	10j	10j	15j	15j	15j	20j	20j	30j	30j
2j	4j	6j	8j	10j	10j	15j	15j	20j	20j	30j	30j	30j	30j	50j	50j	70j
4j	8j	10j	15j	20j	20j	30j	30j	50j	50j	50j	50j	70j	70j	85j	100j	120j



Cardiac: Narrow Complex Tachycardia with Pulse

Pearls:

- Refer to drug formulary charts for all medication dosing for both adults and pediatric patients.
- Use caution in patient currently on antihypertensive medication
- Adenosine may not be effective in identifiable atrial flutter / fibrillation but is not harmful.
- Document all rhythm changes with monitor strips and obtain monitor strips with each therapeutic intervention.
- Continuous pulse oximetry is required for all atrial fibrillation patients.
- Narrow complex tachycardia in setting of alcohol withdrawal should be treated aggressively with midazolam, not diltiazem. If SVT is "exquisitely regular", any heart rate variability should lead you to consider sinus tachycardia or atrial fibrillation.
- Consider a change of vector of initial cardioversion is unsuccessful to anterior/posterior pad placement.
- Sinus tachycardia may be misinterpreted as SVT or A-fib. Sinus tach >150 (adult) or >180 (pediatric) may be seen in the septic patient.